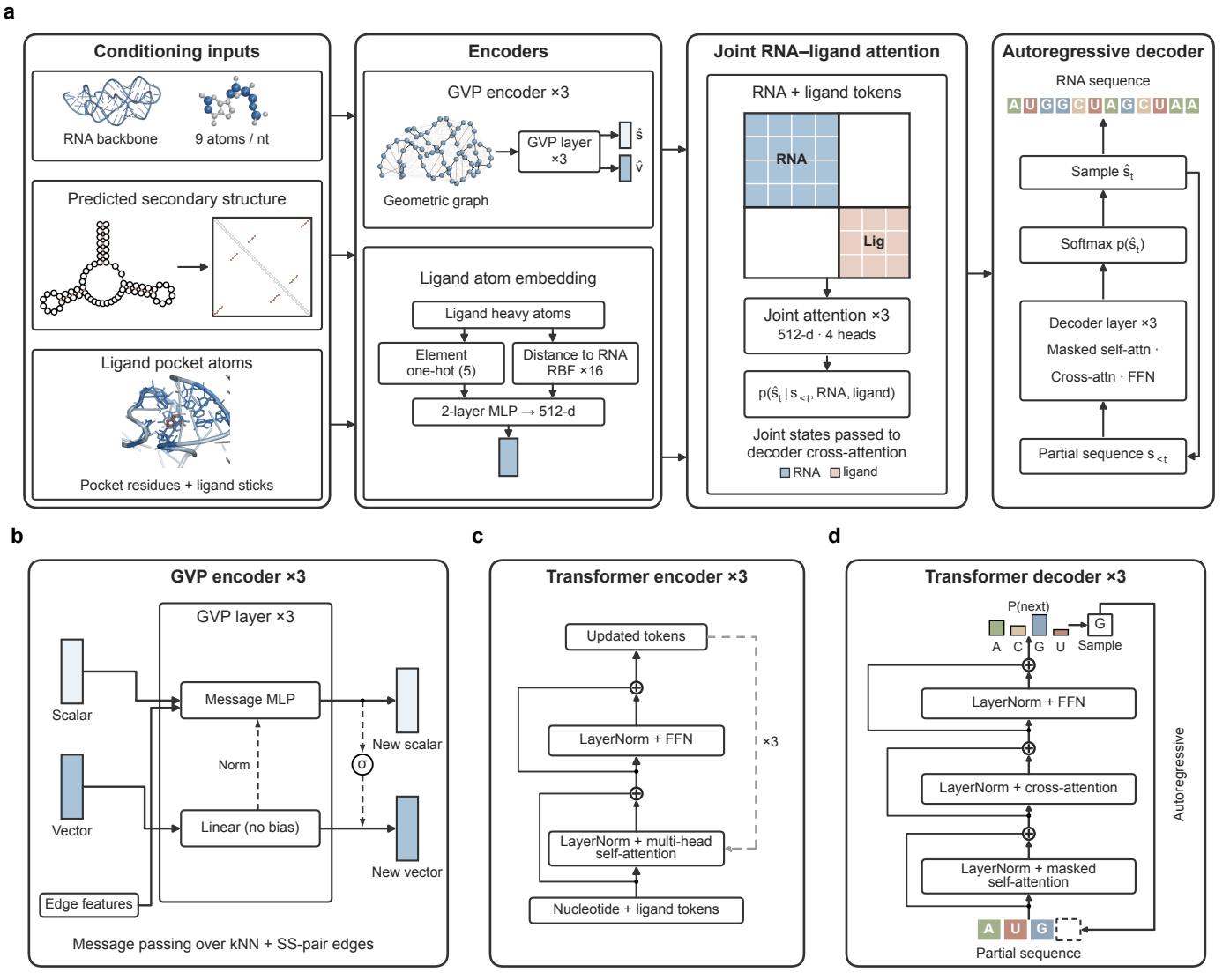

a
Conditioning inputs
RNA backbone
9 atoms / nt
Predicted secondary structure
Ligand pocket atoms
Pocket residues + ligand sticks
Encoders
GVP encoder ×3
Geometric graph
GVP layer ×3
ŝ
v̂
Ligand atom embedding
Ligand heavy atoms
Element one-hot (5)
Distance to RNA RBF ×16
2-layer MLP → 512-d
Joint RNA–ligand attention
RNA + ligand tokens
RNA
Lig
Joint attention ×3
512-d · 4 heads
p(ŝ_t | s_<t, RNA, ligand)
Joint states passed to decoder cross-attention
RNA
ligand
Autoregressive decoder
RNA sequence
A U G G C U A G C U A A
Sample ŝ_t
Softmax p(ŝ_t)
Decoder layer ×3
Masked self-attn · Cross-attn · FFN
Partial sequence s_<t
b
GVP encoder ×3
GVP layer ×3
Scalar
Vector
Message MLP
Norm
Linear (no bias)
σ
New scalar
New vector
Edge features
Message passing over kNN + SS-pair edges
c
Transformer encoder ×3
Updated tokens
LayerNorm + FFN
LayerNorm + multi-head self-attention
Nucleotide + ligand tokens
×3
d
Transformer decoder ×3
P(next)
A C G U
G
Sample
LayerNorm + FFN
LayerNorm + cross-attention
LayerNorm + masked self-attention
Autoregressive
A U G
Partial sequence